

TITLE PAGE

- **Problem Statement ID** –SIH26197

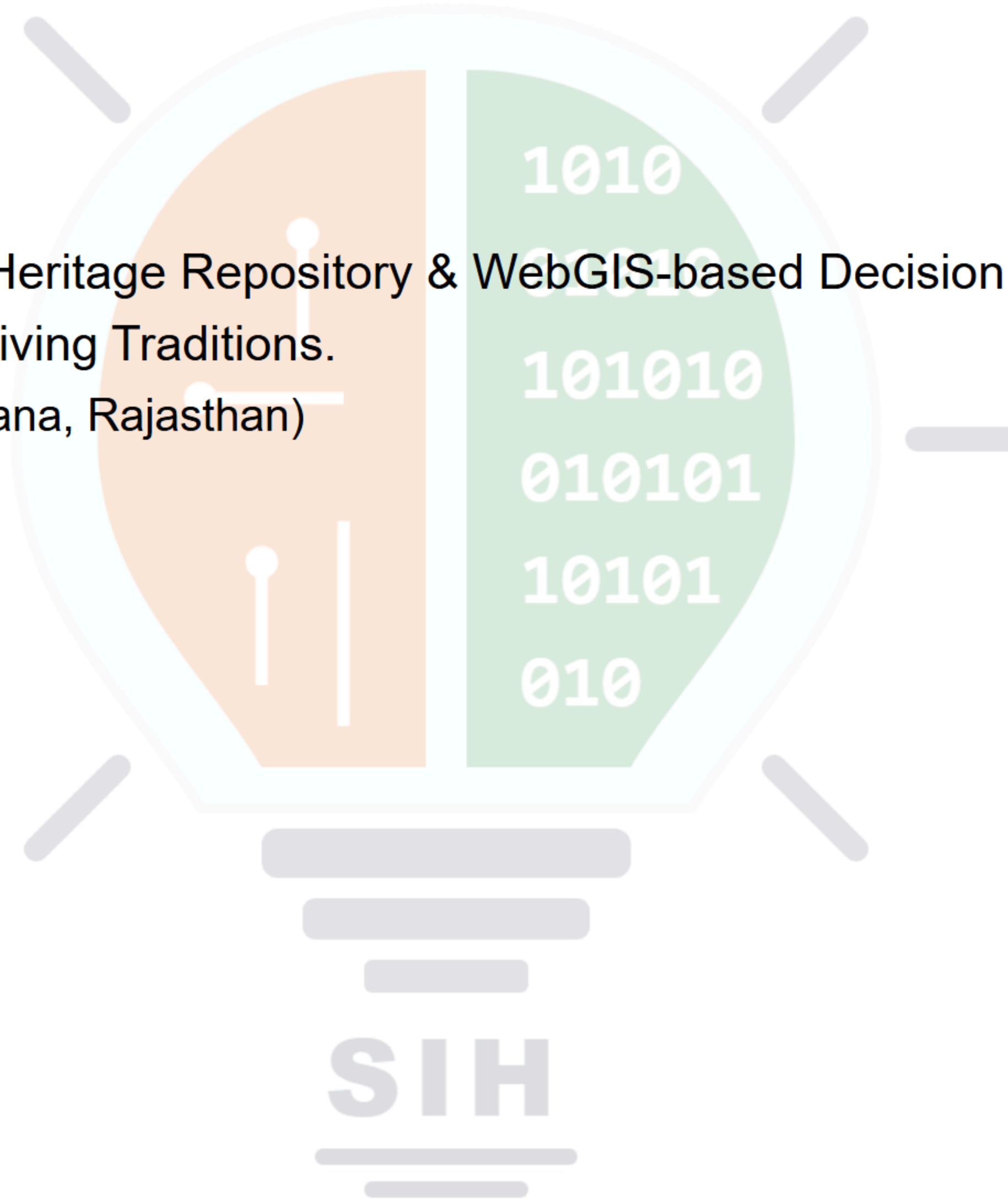
- **Problem Statement Title**-Development of AI-powered Digital Heritage Repository & WebGIS-based Decision Support System (DSS) for Cultural Preservation, Provenance & Living Traditions.
(States to be concentrated: Madhya Pradesh, Tripura , Odisha, Telangana, Rajasthan)

- **Theme**-Miscellaneous

- **PS Category**- Software

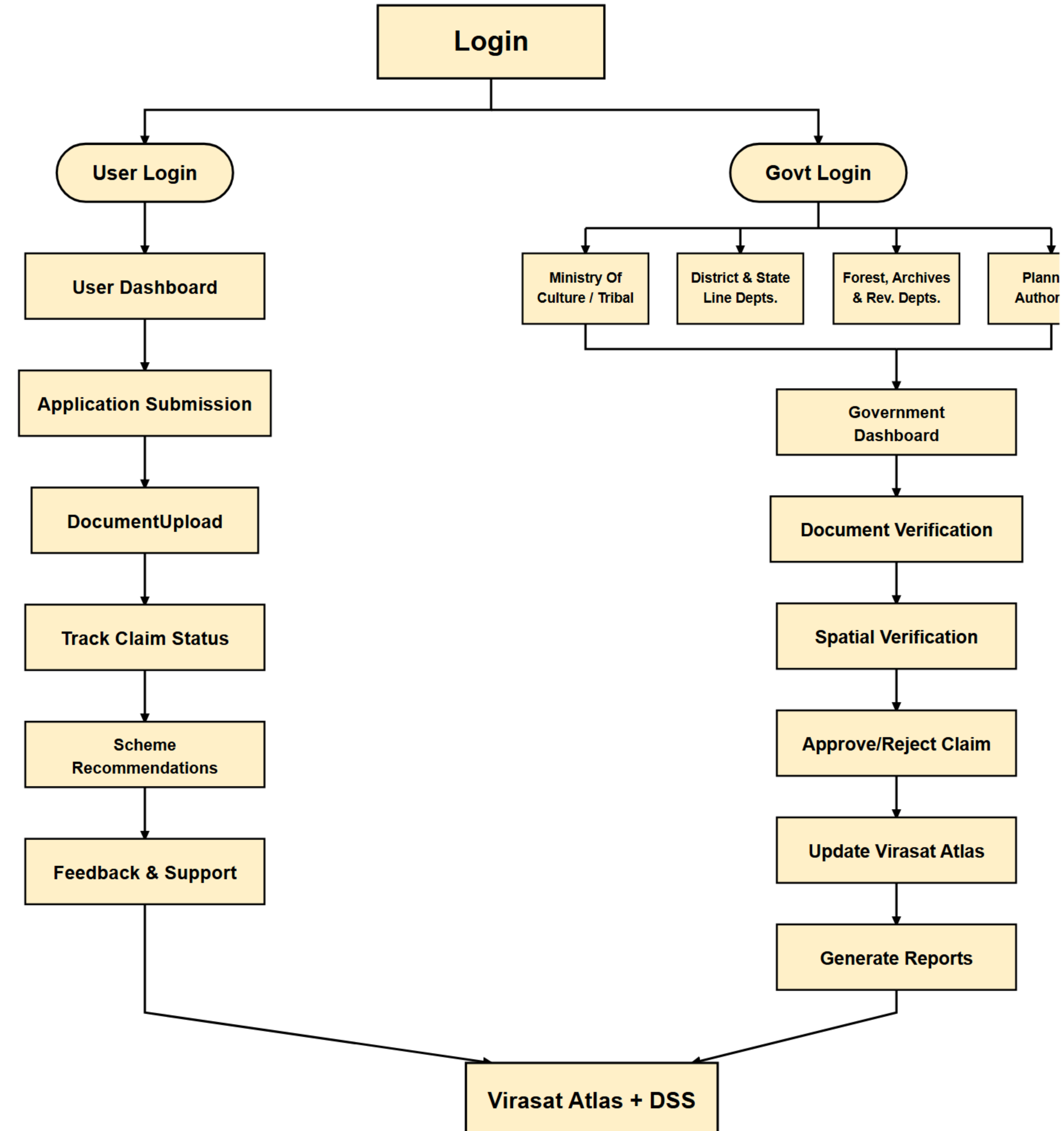
- **Team ID**- 57385

- **Team Name**- Team VirasatX

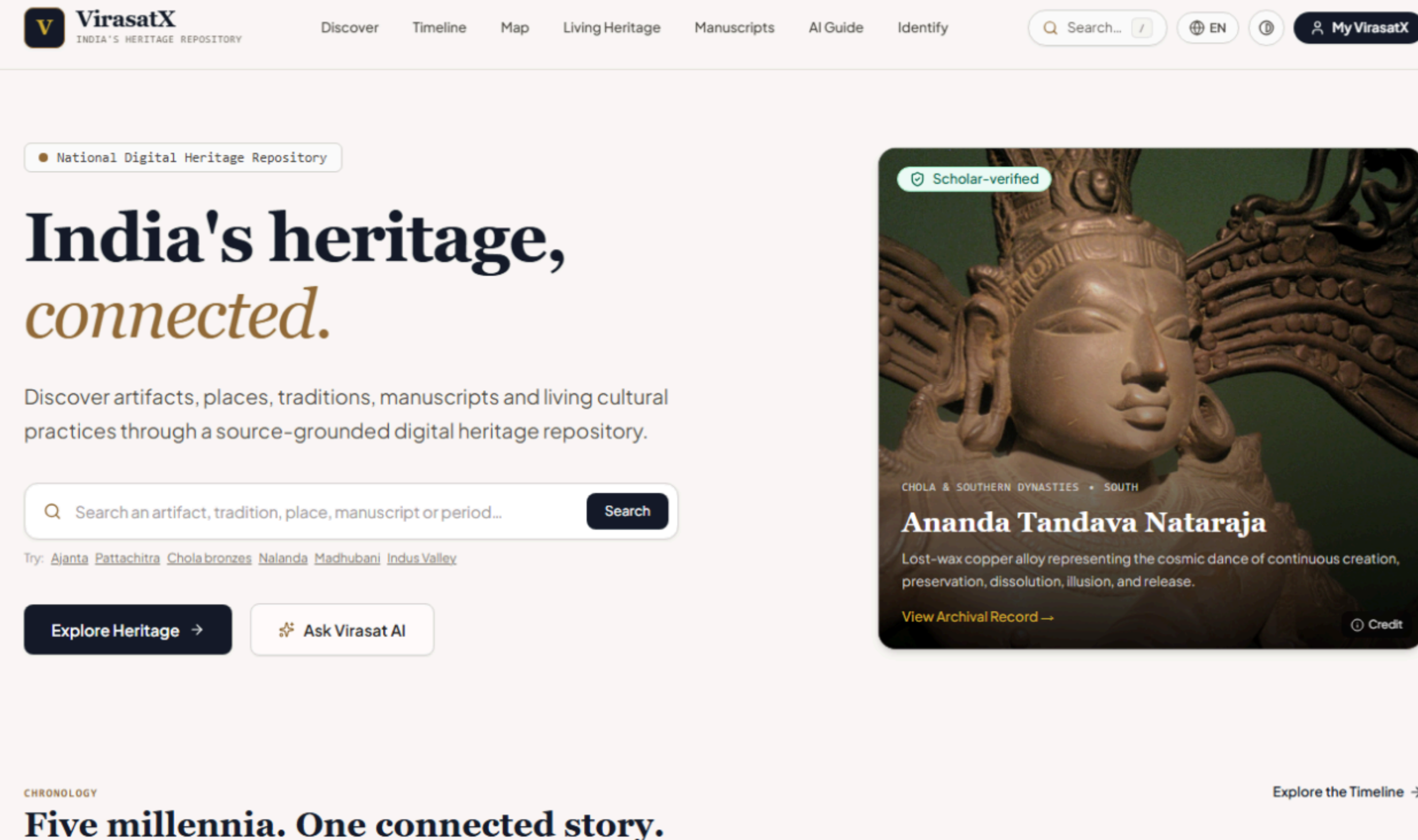
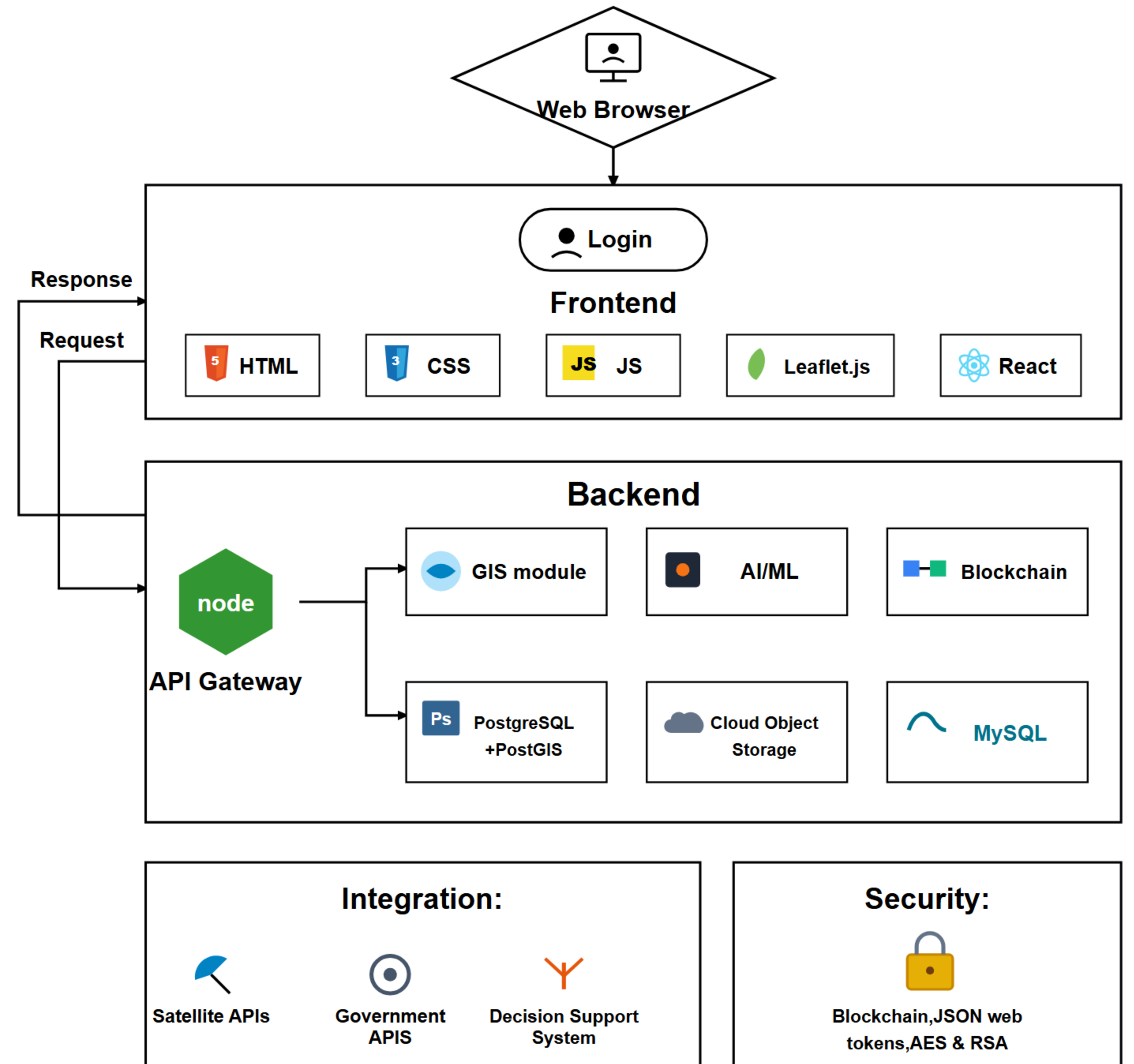


❖ Proposed Solution:

- **Unified Digital Heritage Portal :**
Comprehensive repository linking monuments, artifacts, manuscripts, and living traditions.
- **Digital Witness & Testimony:**
Oral histories, artisan testimonies, and video records stored securely as valid proof.
- **Title Insurance / Digital Patta :**
Ensures authenticity with secure blockchain-backed provenance and artifact registration.
- **Grievance Resolution & Illegal Alerts:**
Efficient heritage grievance handling and unauthorized land encroachment warnings.
- **AI Disciplinary Dashboard :**
Suggests actions and tracks officer and institutional cataloging accountability.
- **Analytics-Driven Officer Monitoring:**
Tracks verification time, claim outcomes, and conservation grievances reported.
- **Interactive Virasat Web-GIS Atlas:**
Dynamic platform for exploring heritage assets, ancient corridors, and excavation sites.
- **AI Disputes Mediator:**
Resolving conflicts and contested claims within existing legal frameworks.
- **GI Tagging and E-Signature :**
To preserve cultural heritage safeguard traditional products and paperless approval .



- **Data Extraction:** Optical Character Recognition(OCR), NLP.
- **WebGIS platform:** Leaflet.js(for interactive maps).
- **Asset Mapping:** Random Forest & CNN.
- **Data Storage:** PostgreSQL & PostGIS(for geospatial data).
- **Data Management:** Cloud Object Storage & MySQL(Structured).
- **GeoServer:** Use standard protocols WMS and WFS to serve maps.
- **Decision Support System:** for scheme recommendations.
- **Blockchain based Security:** Hyperledger Fabric.
- **Key Lock Framework:** Open-source Identity & Access Management
- **Rasa:** Preferred for its offline, multilingual, and secure chatbot.



User Interface: It is a draft version, subject to future updates.

❖ Feasibility:

1. Technical Feasibility:

1. Digitization, AI Mapping and Web-GIS Dashboards:

- OCR and AI extract data from documents & manuscripts.
- Web-GIS dashboards show real-time interactive maps.

2. DSS Integration with Blockchain Security:

- AI engines match schemes to community needs.
- Blockchain ensures secure, tamper-proof digital records.

2. Operational Feasibility:

1. Existing Setup and Simple Staff Training:

- Government data exists, needs proper connection.
- Staff need basic training through short workshops.

2. Stakeholder Support and Effective Heritage Implementation:

- Ministries and NGOs already support cultural preservation schemes.
- System strengthens efforts, not replacing existing work.

3. Legal & Ethical Feasibility:

1. Data Privacy:

- Must follow India's DPDP Act, ensuring legal compliance.
- Blockchain and encryption safeguard sensitive user data.

2. Consent Mechanism:

- Artisan & community informed consent required in all processes.
- Real-time feedback enables secure two-way communication.

❖ Viability:

1. Economic Viability:

1. Cost-Effective & Efficient:

- Saves paperwork, effort, duplication, and reduces fraud.
- Cuts processing delays, expenses, and operational overhead.

2. Supported & Scalable Technology:

- Backed by government programs and tribal inclusion schemes.
- Open-source tech ensures flexible, low-cost, future-ready deployment.

2. Social Viability:

1. Empowers Communities & Improves Scheme Access:

- Provides transparency, secure records, better documentation.
- Directly benefits artisans & custodians through scheme access.

2. Reduces Conflict & Promotes Stronger Inclusion:

- Verified records, alerts reduce frequent attribution disputes.
- Mobile updates, two-way communication improve engagement.

3. Scalability:

1. Pan-India Expansion for Wider Community Benefits:

- Pilot in districts, then scale across India.
- Adjust platform easily for local data.

2. Cross-Sector Use of WebGIS + DSS:

- Adaptable for agriculture, urban development needs.
- Useful in disaster management and planning.

❖ Benefits:

1.Social Benefits:

- Digital platform enables communities with cultural rights tracking.
- IVR helpline, e-learning ensure inclusive literacy access.
- Transparent records, video testimonies reduce conflict, build trust.

2. Environmental Benefits:

- Sustainable site usage by recognized communities & travelers.
- Conservation through community stewardship.
- GI Tagging preserves cultural heritage & traditional products.

3. Economic Benefits:

- Access to schemes (MGNREGA, PM-KISAN, DAGAJU, Jal Jeevan Mission)
- GI Tagging boosts local products & tribal livelihoods
- Title insurance & digital patta ensure secure land ownership → easier loans & income stability

4. Governance Benefits:

- Audit trails ensure accountability, prevent misuse.
- Whistleblowing portal boosts transparency, fights corruption.
- Cross-validation stops duplication and fraudulent claims.

5.Legal & Security Benefits:

- E-Signature & Digital Patta provide legally valid ownership records.
- Protection against land encroachment (Illegal Practice Alert).
- Digital witness/testimony ensures fair dispute settlement.

❖ Impacts:

1.Secure Cultural Rights & Transparency:

- Communities track claims, reducing corruption and fraud.
- Digital patta ensures authenticity in land ownership records.
- Builds trust between citizens and government authorities.

2. Heritage Protection & Conservation with AI:

- AI monitors land to prevent illegal encroachment.
- Conserves biodiversity and safeguards natural resources.
- Promotes sustainable, community-driven cultural preservation.

3. Boost Livelihoods & Local Development:

- Resource planning improves incomes and rural economy.
- GI tagging protects heritage, boosts local products.
- VirasatX enables fair access to welfare schemes.

4. Digital, Eco-Friendly & Cost Effective:

- Reduces paperwork,ensuring greener, eco-friendly governance.
- Preserves tribal knowledge through digital archiving tools.
- Lowers administrative costs and increases efficiency.

5. Smart, Transparent & Scalable Governance:

- Transparent dashboards track officer performance and claims.
- AI dispute mediation ensures fairness, reduces human bias.
- Unified digital platform enables future-ready expansion.

1. AMASR Act & NMM: The Ancient Monuments and Archaeological Sites and Remains Act, 1958 and National Mission on Manuscripts (NMM) provide statutory framework for heritage protection.

<https://asi.nic.in/>

<https://www.namami.gov.in/>

2. Best Practices for Web Security: OWASP Top 10 API Security & Blockchain Verification Standard for Immutable Provenance.

<https://owasp.org/www-project-blockchain-appsec-standard/>

3. The Antiquities and Art Treasures Act: 1972 aims to regulate export trade in antiquities and art treasures, preventing illicit smuggling and unauthorized sales.

<https://asi.nic.in/antiquities-art-treasures-act/>

To enable smooth implementation and integration of the Virasat Atlas System, the following contacts serve as official state-level resources for communication, data coordination, and support:

4. Archaeological Survey of India (ASI): contact.asi@gov.in

5. National Mission on Manuscripts (IGNCA): director.nmm@ignca.nic.in

6. Ministry of Culture, Govt. of India: <https://www.indiaculture.gov.in/>

7. National Portal of India: <https://www.india.gov.in/topics/art-culture>